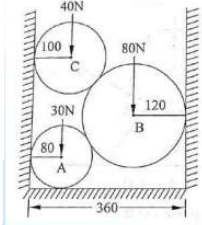


Scheme/ Answer Key for Valuation			
Scheme of evaluation (marks in brackets) and answers of problems/key			
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY			
S1 B.TECH DEGREE (R,S) EXAMINATION, DECEMBER 2020 (2019 Scheme)			
Course Code: EST100			
Course Name: ENGINEERING MECHANICS			
(2019-Scheme)			
Max. Marks: 100			Duration: 3 Hours
PART A			
(Answer all questions, each carries 3 marks.)			
1	statement (1.5), explanation (1.5) of lami's theorem		(3)
2	FBD (1), Example (1), explanation (1)		(3)
3	$\sum F_y = 0$ $N - P \sin 30^\circ - 1000 \cos 30^\circ = 0$ (1 mark) $\sum F_x = 0$ $- 0.25(P \sin 30^\circ + 866.025) - 1000 \sin 30^\circ + P \cos 30^\circ = 0$ (1 mark) $P = (286.06 \text{ N})$ (1 mark)		(3)
4	$\sum M_A = 44 \text{ kN.m (c.w)}$ ----- (1 Mark) Position of R = 6kN (upward) from A, (1 mark) $x = 7.33 \text{ m}$ to the left of A -- (1 mark)		(3)
5	Position vector = $3\hat{i} - 5\hat{j} - 1\hat{k}$ ----- 1 mark Moment = $\vec{r} \times \vec{F} = 9\hat{i} - 1\hat{j} + 32\hat{k}$ ----- 2 mark		(3)
6	Equation – 2 marks, Answer – distance of centre of gravity from O = 140mm. – 1 marks.		(3)
7	$a = -2 \text{ m/s}^2$, (1 marks), Equation (1 mark) Force = -- 100N (1 marks)		(3)
8	Equations & substitution (1+1) Answer (1) --- 53.13°		(3)
9	$A = 0.1157 \text{ m/s}^2$, $\alpha = 0.2315 \text{ rad/s}^2$ (equation (1) + answers 1 marks each)		(3)
10	Differentiate between curvilinear motion and projectile motion. 1.5 + 1.5		(3)
PART B			
(Answer one full question from each module, each question carries 14 marks)			
Module-I			
11	a) Free body diagram - (1 marks) $\Theta = 61.04^\circ$ & $\Phi = 43.43^\circ$ - (1 marks) Equilibrium equations - (1 marks) $W = 572.66 \text{ N}$ - (1 marks) $T_{CA} = 619.58 \text{ N}$, $T_{CD} = 542.11 \text{ N}$, & $T_{DB} = 788.56 \text{ N}$ - (1 marks)		(5)

	b)	$\sum F_H = R \rightarrow P \cos \Phi = -8.284$ - (3 marks) $\sum F_V = 0 \rightarrow P \sin \Phi = 2.296$ - (3 marks) $\Phi = 15.5^\circ$ - (1 marks) $P = 8.60 \text{ kN}$ - (2 marks)	(9)
12	a)	Three cylinders are piled in a rectangular ditch as in figure. Neglecting friction, determine the reaction between cylinder A and vertical wall. FBD of A, B, C – (2 marks each) $R_{BC} = 51.85 \text{ N}$, (2 marks) angle of BC with Horizontal $= 50.48^\circ$ (1 mark) $R_{AB} = 200 \text{ N}$, (2 marks) angle of BC with Horizontal $= 36.87^\circ$ (1 marks) $R_{wall} = 160 \text{ N}$ (2 marks) 	(14)
Module-II			
13	a)	$\sum F_y = 0$; $R_{AV} + R_B \sin 53.13 - 10 = 0 \rightarrow$ 1 marks $\sum F_x = 0$; $R_{AH} + 4 - 4 - R_B \cos 53.13 = 0 \rightarrow$ 1 marks $\sum M_A = 0$; $-10 \times 2 - 8 + R_B \sin 53.13 \times 6 = 0 \rightarrow$ 1 marks $\tan \Theta = \frac{3}{4}$; $\Theta = 36.87^\circ \rightarrow$ 1 marks $R_B = 5.83 \text{ kN}$ ----- 1 marks $R_{AV} = 5.33 \text{ kN}$ & $R_{AH} = 3.5 \text{ kN}$ $R_A = 6.38 \text{ kN}$ ----- 1 marks $\theta_A = 33.29^\circ$ with vertical ----- 1 mark	(7)
	b)	FBD of ladder - 1 mark, $\sum F_x = 0$; $N_w - 0.4 \times N_f = 0$ $\sum F_y = 0$; $N_f + 0.6 \times N_w - 98 = 0$ (Θ , inclination of ladder with floor) $N_f = 79.03 \text{ N}$; $N_w = 31.61 \text{ N}$ - 1.5 + 1.5 marks, $\sum M_{\text{wrt floor}} = 0$ - 1 mark, $98 \times l/2 \cos \Theta - N_w \times l \sin \Theta - 0.6 \times N_w - l \cos \Theta$ (l is length of ladder) $\Theta = 43.53^\circ$ - 2 mark,	(7)
14	a)	F.B.D of A ----- 2 marks F.B.D of B ---- 2marks Equations of equilibrium for block A $\sum F_x = 0$: $N_A - f_{AB} \cos 10^\circ - N_{AB} \sin \theta = 0$ ----- 2 marks $\sum F_y = 0$: $-f_A - f_{AB} \sin 10^\circ + N_{AB} \cos \theta - 2.943 \text{ kN} = 0$ ----- 2 marks Equations of equilibrium for block B $\sum F_x = 0$: $f_{AB} + (1.115 \text{ N}) \cos 10^\circ + (3.716 \text{ N}) \sin 10^\circ - P = 0$ - 2 marks $\sum F_y = 0$: $N_B + (1.115 \text{ N}) \sin 10^\circ - (3.716 \text{ N}) \cos 10^\circ = 0$ ----- 2 marks Solving for $P = 2780 \text{ N}$ ----- 2 marks	(14)

Module-III			
15		$a_1 = 1200 \text{ mm}^2$, $a_2 = 7850 \text{ mm}^2$, $a_3 = 2400 \text{ mm}^2$, $a_3 = 28000 \text{ mm}^2$, $x_1 = 10 \text{ mm}$, $x_2 = 100 \text{ mm}$, $x_3 = 180 \text{ mm}$, $x_4 = 100 \text{ mm}$, $y_1 = 113.3 \text{ mm}$, $y_2 = 70 \text{ mm}$, $y_3 = 110 \text{ mm}$, $y_4 = 70 \text{ mm}$ ----- 12 marks (above values depends upon how the total area is divided) $\bar{x} = 94.9 \text{ mm}$ and $\bar{y} = 61.1 \text{ mm}$ ----- 2+2 marks	(14)
16		$A(0,12,0)$ $B(-4,0,-3)$ $C(-4,0,6)$ $D(6,0,4)$ (4 marks) Force vectors (6) Resultant force = 674.45N (4)	
Module-IV			
17	a)	Constraint equation string gives : acceleration of block A downwards = 2 x acceleration of pulley Q upwards (2 mark) FBD of block A (1 mark) FBD of block B (1 mark) FBD of Pulley Q (1 mark) Equation of motion of block A (1 mark) Equation of motion of block B (1 mark) Equation of motion of Pulley Q (1 mark) Acceleration of block A = 2.08 m/s^2 (2 marks) Acceleration of block B = 1.04 m/s^2 (2 marks) Tension in string = 542.56 N (2 marks)	(14)
18	a)	$e = 0.816$, $H = 13.5 \text{ m}$. $H_1 = e^2 H_0$; $H_2 = e^2 H_1$; $e = 0.8165$ (1 + 1 + 1 marks) $H_0 = 13.5 \text{ m}$ (2 marks)	(5)
	b)	B overtake at 10.04 s (4 marks) Speed of A (7.53 m/s) (1 marks) Speed of B (9.12 m/s) (1 marks) Distance at which B OVERTAKE (37.8 m) (2 marks)	(9)
Module-V			
19		Ans: a) $t = 1.26 \text{ sec}$ b) velocity = 4.75 m/s Equation of motion of the drum... 2 marks MI of the drum... .2 marks Equation of motion of the weight Q... 2 marks Expression for tension in the cord... 2 marks Tangential acceleration calculation from equations of motion ... 2 mark Velocity calculation 2 marks Time calculation... 2 marks	(14)
20	a)	Stiffness in N/m = 800 N/m... (1) $\omega = 12.52 \text{ rad/s}$ Time period = $2\pi / \omega = 0.5 \text{ s}$... (2) Expression for time period... (2)	(5)
	b)	$v = \omega \sqrt{(r^2 - x^2)}$ + Substitution (1 + 1 marks) $r = 0.252 \text{ m}$, $\omega = 5.218 \text{ rad/s}$, $t = 1.218 \text{ seconds}$, (1 + 1 + 1 MARKS) $V_{\max} = 1.30 \text{ m/s}$, $a_{\max} = 6.74 \text{ m/s}^2$ (2 + 2 marks)	(9)
